

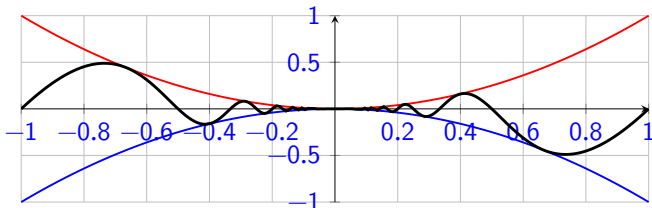
2627-MA140: Week 02, Lecture 3 (L06)

Limits; ~~The Squeeze Theorem~~

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Thursday, 17 September, 2026



Topics we'll squeeze into today's class:

- 1 Recall... Limits
 - Properties of Limits
 - Rational functions
- 2 Limits of Rational Functions
- 3 Completing the square
- ~~4 The Squeeze Theorem~~
- ~~5 $\sin(\theta)/\theta$~~
 - ~~■ Other examples~~
- 6 Exercises

For more, see Chapter 2 (Limits) of Strang and Herman's **Calculus**, especially Section and 2.3 (Limit Laws).

Slides are on canvas, and at niallmadden.ie/2627-MA140

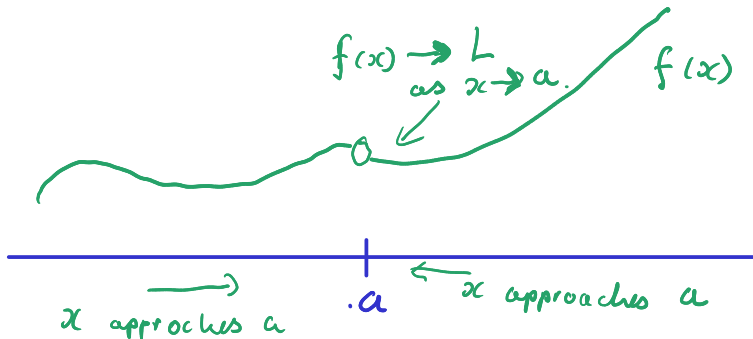


Recall... Limits

Yesterday, we learned that

$$\lim_{x \rightarrow a} f(x) = L,$$

means that we can make $f(x)$ as close to L as we like, by taking x as close to a as needed.



We learned about the following “Limit Laws”: *Suppose that*

$$\lim_{x \rightarrow a} f_1(x) = L_1 \quad \text{and} \quad \lim_{x \rightarrow a} f_2(x) = L_2,$$

and $c \in \mathbb{R}$ *is any constant.* Then,

$$(1) \quad \lim_{x \rightarrow a} c = c, \quad c \in \mathbb{R}$$

$$(2) \quad \lim_{x \rightarrow a} x = a$$

$$(3) \quad \lim_{x \rightarrow a} [c f_1(x)] = c L_1$$

$$(4) \quad \lim_{x \rightarrow a} [f_1(x) + f_2(x)] = L_1 + L_2$$

and

$$\lim_{x \rightarrow a} [f_1(x) - f_2(x)] = L_1 - L_2$$

$$(5) \quad \lim_{x \rightarrow a} (f_1(x) f_2(x)) = L_1 L_2$$

$$(6) \quad \lim_{x \rightarrow a} ((f_1(x))^n) = (L_1)^n$$

$$(7) \quad \lim_{x \rightarrow a} \left(\frac{f_1(x)}{f_2(x)} \right) = \frac{L_1}{L_2},$$

providing $L_2 \neq 0$.

$$(8) \quad \lim_{x \rightarrow a} \sqrt[n]{f_1(x)} = \sqrt[n]{L_1}$$

In many cases, evaluating limits is more complicated. In particular, we'll consider numerous examples where we want to evaluate $\lim_{x \rightarrow a} f(x)$ where a is not in the domain of f .

A typical example of this is when we evaluate a rational function,

$\lim_{x \rightarrow a} \frac{p(x)}{q(x)}$ where **both** $p(a) = 0$ and $q(a) = 0$.

Since we care about the value of p and q **near** $x = a$, but not actually at $x = a$, we said is safe to factor out an $(x - a)$ term from both.

We finished with showing that

$$(a) \lim_{x \rightarrow 0} \frac{x}{x} = 1 \quad (b) \lim_{x \rightarrow 0} \frac{x^2}{x} = 0$$

$$(c) \lim_{x \rightarrow 0} \frac{x}{x^2} \text{ is not defined}$$

Limits of Rational Functions

When trying to evaluate $\lim_{x \rightarrow a} \frac{p(x)}{q(x)}$, where $p(x)$ and $q(x)$ are both polynomials, of the same degree...

- ▶ Check that $p(a) = 0$ and $q(a) = 0$;
- ▶ If so, we should be able to factor $(x - a)$ from both.
- ▶ Do that, and so simplify $\frac{p(x)}{q(x)}$ (for $x \neq a$, of course).
- ▶ Try to evaluate the limit of the resulting simpler rational function.

Recall that the degree of a polynomial is the highest power of x . Eg

$$\deg(x^2) = 2 \qquad \deg(1 + 3x - 52x^4 + x^2) = 4$$

Limits of Rational Functions

Example

Evaluate the limit

$$\lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x^2 - x} = \lim_{x \rightarrow 1} \frac{p(x)}{q(x)}$$

Let $p(x) = x^2 + x - 2$

$q(x) = x^2 - x$

Check $p(1) = 1 + 1 - 2 = 0$

$q(1) = 1 - 1 = 0$



So we divide both by $(x-1)$.

$$p(x) = (x^2 + x - 2) = (x-1)(x+2)$$

$$q(x) = x^2 - x = (x-1)x$$

$$\text{So } \frac{p(x)}{q(x)} = \frac{\cancel{(x-1)}(x+2)}{\cancel{(x-1)}x} = \frac{x+2}{x} \text{ for } x \neq 1.$$

$$\text{So } \lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x^2 - x} = \lim_{x \rightarrow 1} \frac{x+2}{x} = \frac{3}{1} = 3$$



Limits of Rational Functions

In that last example, we found that

$$\lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x^2 - x} = \lim_{x \rightarrow 1} \frac{x + 2}{x}$$

But these are different functions:

But $\frac{x^2 + x - 2}{x^2 - x}$ is not the same
function as $\frac{x + 2}{x}$.

Because they have different domains (why?).

Limits of Rational Functions

Evaluate the limit

$$\lim_{x \rightarrow 2} \left(\frac{\frac{1}{2} - \frac{1}{x}}{x - 2} \right) = \lim_{x \rightarrow 2} \frac{p(x)}{q(x)}$$

$$p(x) = \frac{1}{2} - \frac{1}{x} = \frac{x}{2x} - \frac{2}{2x} = \frac{x-2}{2x}$$

$$q(x) = x - 2$$

$$p(2) = \frac{1}{2} - \frac{1}{2} = 0 \checkmark$$

$$q(2) = 2 - 2 = 0 \checkmark$$

So now $\frac{p(x)}{q(x)} = p(x) \frac{1}{q(x)} = \frac{x-2}{2x} \cdot \frac{1}{x-2} = \frac{1}{2x}$ for $x \neq 2$.

$$\text{Then } \lim_{x \rightarrow 2} \frac{p(x)}{q(x)} = \lim_{x \rightarrow 2} \frac{1}{2x} = \boxed{\frac{1}{4}} \checkmark$$

Completing the square

Very often, we'll evaluate limits of the form:

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$$

where f and g are not polynomials. Some of the same ideas still apply.

Example

Evaluate

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x^2} - 1}{x^2} = \lim_{x \rightarrow 0} \frac{p(x)}{q(x)}$$

check $p(0) = \sqrt{1+0^2} - 1 = 1 - 1 = 0$ ✓
 $q(0) = 0^2 = 0$ ✓

Completing the square

We have

$$\frac{\sqrt{1+x^2} - 1}{x^2} = \frac{(\sqrt{1+x^2} - 1)(\sqrt{1+x^2} + 1)}{x^2(\sqrt{1+x^2} + 1)}$$

$$= \frac{(\sqrt{1+x^2})(\sqrt{1+x^2}) - \sqrt{1+x^2} + \sqrt{1+x^2} - 1}{x^2(\sqrt{1+x^2} + 1)}$$

$$= \frac{(1+x^2) + 0 - 1}{x^2(\sqrt{1+x^2} + 1)} = \frac{x^2}{x^2(\sqrt{1+x^2} + 1)}$$

$$= \frac{1}{\sqrt{1+x^2} + 1} \quad x \neq 0.$$

$$\text{So } \lim_{x \rightarrow 0} \frac{p(x)}{q(x)} = \lim_{x \rightarrow 0} \frac{1}{\sqrt{1+x^2} + 1} = \boxed{\frac{1}{2}} \quad \checkmark$$

The Squeeze Theorem

There are various approaches to evaluating limits, including...

The Squeeze Theorem (a.k.a. Sandwich Theorem)

Suppose that we have three functions f , g and h on some interval $[x_0, x_1]$, with

$$g(x) \leq f(x) \leq h(x),$$

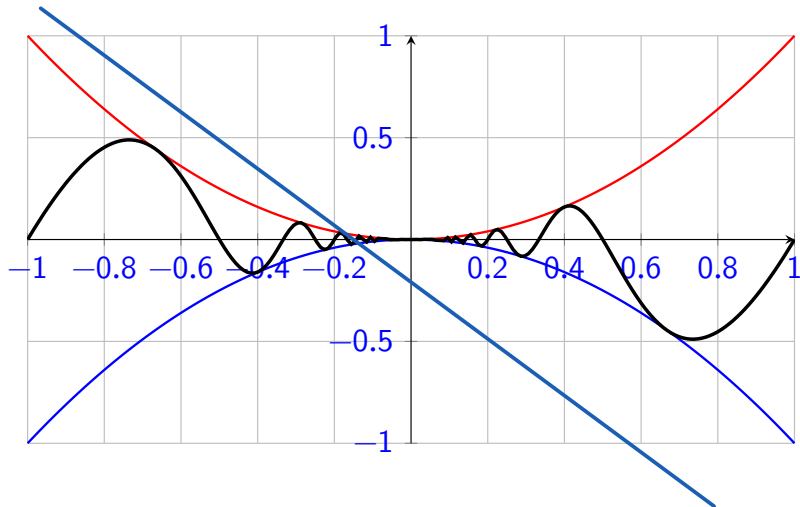
and

$$\lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} h(x) = L,$$

for some $a \in [x_0, x_1]$. Then $\lim_{x \rightarrow a} f(x) = L$.

That is: if $g(x)$ and $h(x)$ have the same limit as $x \rightarrow a$, and $f(x)$ is “squeezed” between them, then $f(x)$ has that same limit too as $x \rightarrow a$.

The Squeeze Theorem



The Squeeze Theorem

Example

Suppose $f(x)$ is a function such that

$$1 - \frac{x^2}{4} \leq f(x) \leq 1 + \frac{x^2}{2}, \quad \forall x \neq 0$$

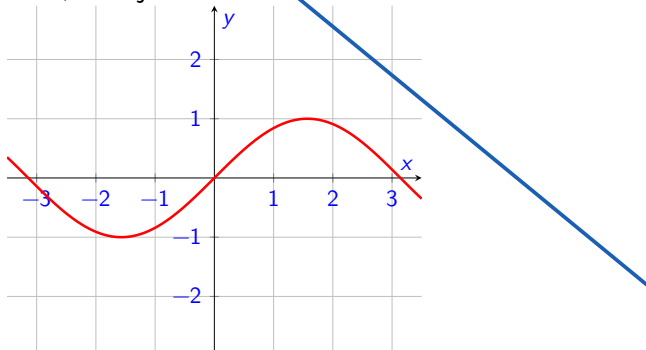
Find $\lim_{x \rightarrow 0} f(x)$.

$$\sin(\theta)/\theta$$

Next, we will use the Squeeze Theorem to explain **an important limit**:

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$$

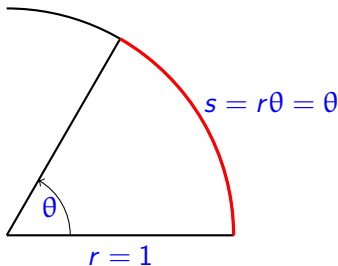
First, let's just convince ourselves:



Next, we review some facts about trigonometric functions.

- ▶ **In this module, we only every use radians** (never degrees).
- ▶ Given a right-angled triangle, with sides a , b and h (the hypotenuse), $\sin \theta = \frac{b}{h}$, $\cos \theta = \frac{a}{h}$, $\tan \theta = \frac{b}{a} = \frac{\sin \theta}{\cos \theta}$

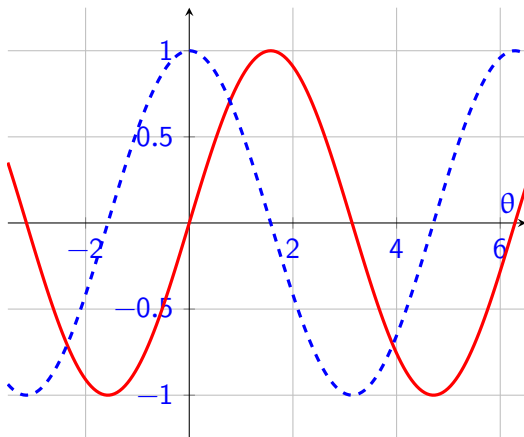
- ▶ The length of a sector, subtended by the angle θ , of a circle of radius r , is $s = r\theta$. In particular, for the unit circle **the angle (in radians) is the length of the arc..**



- ▶ Area of a sector of a circle is $\frac{1}{2}r^2\theta$ where r is the radius of the circle, and θ is the angle subtended by the sector.

$$\sin(\theta)/\theta$$

► The $\sin$ (red) and $\cos$ (blue) functions look like this:



Various other facts are summarised in the State Examination Commission's Tables:

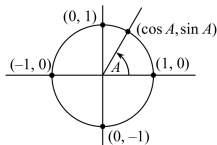
Triantánacht

$$\tan A = \frac{\sin A}{\cos A}$$

$$\cot A = \frac{\cos A}{\sin A}$$

$$\sec A = \frac{1}{\cos A}$$

$$\operatorname{cosec} A = \frac{1}{\sin A}$$



Trigonometry

$$\cos^2 A + \sin^2 A = 1$$

$$\sec^2 A = 1 + \tan^2 A$$

$$\cos(-A) = \cos A$$

$$\sin(-A) = -\sin A$$

$$\tan(-A) = -\tan A$$

Nóta: Bíonn $\tan A$ agus $\sec A$ gan sainiú nuair $\cos A = 0$.
Bíonn $\cot A$ agus $\operatorname{cosec} A$ gan sainiú nuair $\sin A = 0$.

Note: $\tan A$ and $\sec A$ are not defined when $\cos A = 0$.
 $\cot A$ and $\operatorname{cosec} A$ are not defined when $\sin A = 0$.

A (céimeanna)	0°	90°	180°	270°	30°	45°	60°	A (degrees)
A (raidian)	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	A (radians)
$\cos A$	1	0	-1	0	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	$\cos A$
$\sin A$	0	1	0	-1	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	$\sin A$
$\tan A$	0	-	0	-	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	$\tan A$

$$1 \text{ rad.} \approx 57.296^\circ$$

$$1^\circ \approx 0.01745 \text{ rad.}$$

Foirmlí uillinneacha comhshuite

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

Compound angle formulae

$$\cos(A-B) = \cos A \cos B + \sin A \sin B$$

$$\sin(A-B) = \sin A \cos B - \cos A \sin B$$

$$\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

Foirmlí uillinneacha dúbailte

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos^2 A = \frac{1}{2}(1 + \cos 2A)$$

$$\sin^2 A = \frac{1}{2}(1 - \cos 2A)$$

Double angle formulae

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$$\cos 2A = \frac{1 - \tan^2 A}{1 + \tan^2 A}$$

$$\sin 2A = \frac{2 \tan A}{1 + \tan^2 A}$$

Iolraigh a thiontú ina suimeanna agus ina ndifríochtaí**Products to sums and differences**

$$2 \cos A \cos B = \cos(A + B) + \cos(A - B)$$

$$2 \sin A \cos B = \sin(A + B) + \sin(A - B)$$

$$2 \sin A \sin B = \cos(A - B) - \cos(A + B)$$

$$2 \cos A \sin B = \sin(A + B) - \sin(A - B)$$

Suimeanna agus difríochtaí a thiontú ina n-iolraigh**Sums and differences to products**

$$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$$

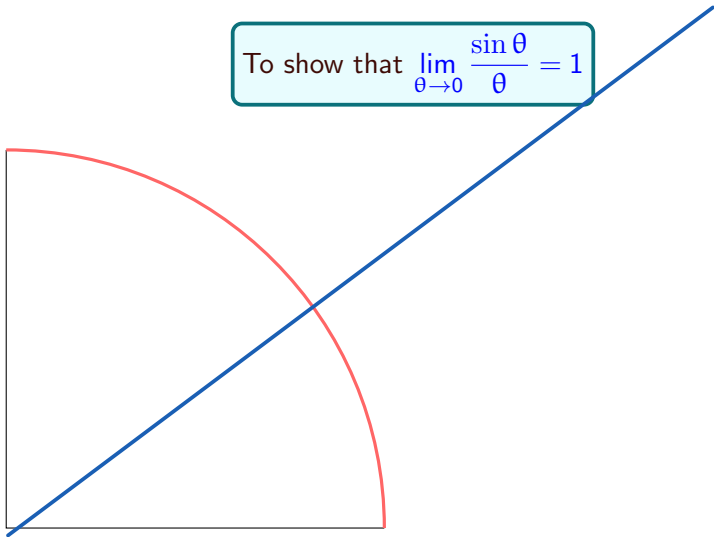
$$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\sin(\theta)/\theta$$

To show that $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$



$$\sin(\theta)/\theta$$

Example

Evaluate $\lim_{x \rightarrow 0} \frac{\tan(3x)}{\sin(2x)}$

Example

Evaluate $\lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta^2}$

Exercise 2.3.1

Evaluate the following limits

$$(a) \lim_{x \rightarrow \frac{1}{2}} \frac{x - \frac{1}{2}}{x^2 - \frac{1}{4}}$$

$$(b) \lim_{x \rightarrow -4} \frac{x^2 + 3x - 4}{x^2 + x - 12}$$

Exercise 2.3.2

(From 2023/2024 MA140 exam, Q1(a)) Evaluate the limit

$$\lim_{x \rightarrow 4} \frac{x - 4}{(\sqrt{x} - 2)(x + 9)}$$

Exercises

Exercise 2.3.3

Suppose that $g(x) = 9x^2 - 3x + 1/4$, and $f(x)$ is such that $-g(x) \leq f(x) \leq g(x)$ for all x .

1. Can one use the Squeeze Theorem to determine $\lim_{x \rightarrow 1/3} f(x)$? If so, do so. If not, explain why.
2. Can one use the Squeeze Theorem to determine $\lim_{x \rightarrow 1/6} f(x)$? If so, do so. If not, explain why.

Exercises

Exercise 2.3.4 (from 2425-MA140 exam)

Let $f(x) = \frac{x^2 - 2x - 15}{3x^3 - 6x^2 - 45x}$. For each of the following, evaluate the limit, or determine that it does not exist.

(i) $\lim_{x \rightarrow -3} f(x)$

(ii) $\lim_{x \rightarrow 0} f(x)$

(iii) $\lim_{x \rightarrow 5} f(x)$

Exercises

Exercise 2.3.5 (from 2425-MA140 Exam)

Suppose that $g(x) = 2x^4 + x^2$, and $f(x)$ is such that $-g(x) \leq f(x) \leq g(x)$ for all x .

1. Can one use the Squeeze Theorem to determine $\lim_{x \rightarrow 0} f(x)$? If so, do so. If not, explain why.
2. Can one use the Squeeze Theorem to determine $\lim_{x \rightarrow 1} f(x)$? If so, do so. If not, explain why.